Luiz Cauet
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EDUCATION

University of California, Santa Cruz — Santa Cruz, California

B.S. in Computer Engineering (Systems Programming) CS Minor — *Expected Graduation, March 2027*

GPA: 3.5

Related Coursework: Data Structures & Algorithms, Networks, Systems Design, Computer Architecture, Intro to Software Engineering, Computer Organization, Game AI

EXPERIENCE

Prosper XO — Austin, TX

Software Engineering Intern — *July 2025 - Sep 2025*

- Developed features for a React + Supabase music-tech web application
- Built reusable UI components and implemented backend database integrations
- Prototyped product features and contributed to technical planning and design discussions

Augmented Design Lab — Santa Cruz, California

Research Participant (Advisor: Dr. Jim Whitehead) — *Jan 2026 – Current*

- Developed PCG prototypes using LLMs, Wave Function Collapse, and Z3 constraint solving
- Implemented grid-based world generation systems and constraint-driven content pipelines
- Analyzed and applied technical research in procedural generation and game systems

The Slug Collective — Santa Cruz, California

Founder & President — *Dec 2024 – Current*

- Founded and led a 100+ member interdisciplinary music & tech collective
- Secured $1,500 in funding through technical project proposals and organizational leadership
- Directed collaborative production of a 12-track multi-contributor digital release

PROJECTS

Resonance - Text-to-Synth Preset Generator (ML, LLM, Python, Next.js) — *2026 - Current*

- Developed a multimodal ML pipeline mapping natural language prompts to synthesizer presets using CLAP audio-text embeddings and vector similarity search
- Experimented with LLM-driven iterative fine-tuning of preset parameters based on user feedback
- Implemented Python + FastAPI backend services for embedding generation, retrieval, and preset manipulation; containerized ML pipelines on Oracle Cloud VM with Docker.
- Built a Next.js interface enabling real-time semantic search and interactive preset modification

SmartQueue (Next.js, React, TypeScript, Supabase, Spotify SDK) — *2025*

- Built an adaptive queue system that dynamically reorders playlists based on user behavior
- Designed a weighted recommendation algorithm with persistent canonical state using Supabase Auth and RLS

Music Collaboration Forum (Node.js/Express.js, AWS S3, React) — *2024*

- 1st place hackathon winner. Built a full-stack collaboration platform supporting authentication, media uploads, and real-time user interaction
- Implemented AWS S3-backed audio storage and scalable backend services

SKILLS

- **Languages:** Python, JavaScript/TypeScript, C/C++, Java, SQL
- **Web & Backend:** React, Next.js, Node.js, Express, FastAPI, Supabase
- **Systems & Tools:** Git, Docker, Unix/Linux, AWS, Oracle Cloud, Wireshark
- **Game & Creative Tech:** Godot, Audio Processing, Sound Design, FL Studio
- **Languages Spoken:** English, Portuguese, Spanish